Comparative Study of Multifunctional Biodegradable Bio-Coagulant Beads for Smart Water Treatment Applications

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INTRODUCTION:

Turbidity: Turbidity is an important parameter used to determine water contamination levels. High turbidity affects water clarity, blocks light penetration, and may indicate the presence of harmful pollutants and microorganisms.

Limitations of Conventional System: Conventional purification systems are often expensive, chemical-intensive, and less sustainable for low-resource applications.

The Need for Innovation: Therefore, there is a growing need for eco-friendly, biodegradable, and low-cost water treatment technologies.

PROBLEM STATEMENT:

- **Rising Contamination:**

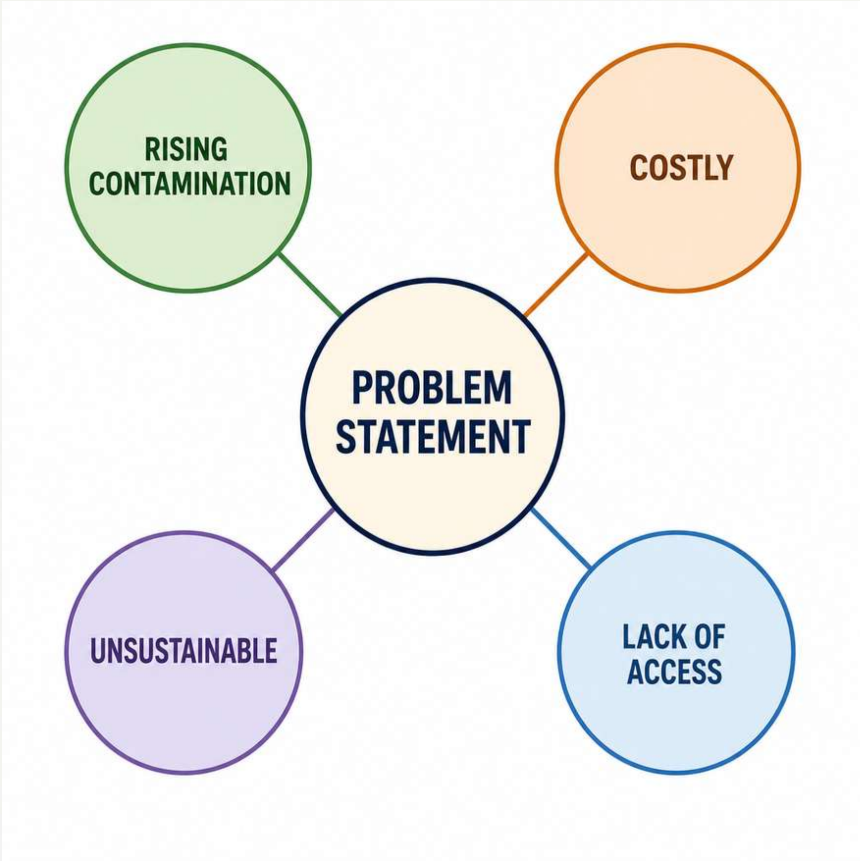
Water contamination due to suspended particles and pollutants is increasing globally.

- **Costly & Unsustainable:**

Conventional purification systems can be expensive and environmentallyunsustainable.

- **Lack of Access:**

Conventional purification systems can be expensive and environmentally unsustainable.



AIM OF THE EXPERIMENT:

- Reducing Turbidity
- Smart Water Quality Monitoring
- Improving Water Clarity
- Enhancing Coagulation & Adsorption

CORE CONCEPT:

The project utilizes natural biomaterials encapsulated within sodium alginate hydrogel beads to combine three powerful mechanisms:-

Coagulation

Neutralization and settling of suspended particles.

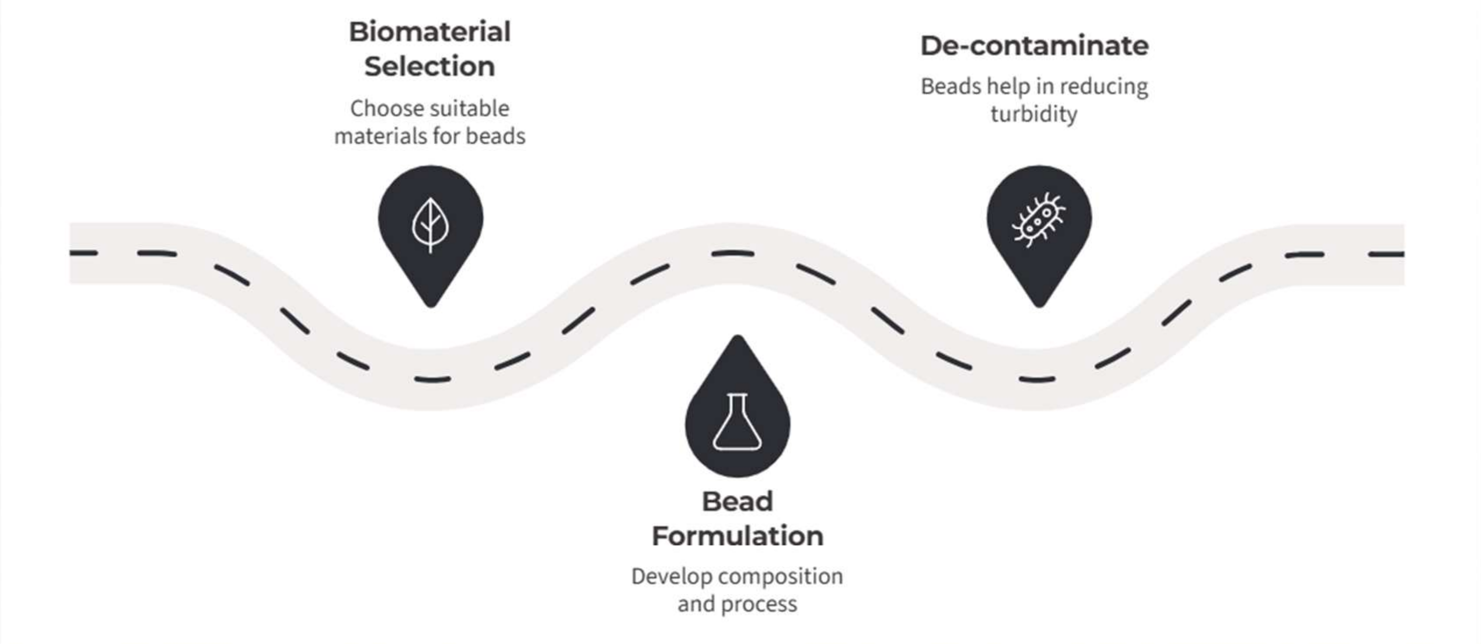
Adsorption

Binding and removal of impurities and contaminants.

Smart Monitoring

Real-time turbidity analysis using sensors integrated with ESP32.

THE BIOLOGY INTEGRATION



WHY MORINGA?

Positively charged water-soluble proteins	
2S albumin protein	**MOCP**

- Neutralizes negatively charged suspended particles in dirty water
- Floc formation and coagulation


WHY BANANA PEEL?

Hemicellulose	**Pectin**
Cellulose	**Lignin**

- Improves biosorption
- Enhances contaminant binding
- Increases adsorption efficiency


WHY CHARCOAL?

Porous surfaces | **Adsorption sites**

- Helps in trapping pollutants, organic impurities, odor molecules and some metal ions.

WHY ORANGE PEEL?

Pectin | **Carboxyl group**

- Adsorption enhancement
- Pollutant interaction
- Improved purification efficiency

HOW WE MADE THE BEADS
Step-by-Step Process (Final Bead – with All Powders)
1 RAW MATERIALS COLLECTION
Collected moringa seeds, banana peels, orange peels, and charcoal.
2 SLIGHT DE-MOISTING
All materials were washed and naturally de-moistened to remove excess moisture.
3 GRINDING INTO FINE POWDERS
Dried materials were ground separately into fine powders using a grinder.
4 PREPARING SODIUM ALGINATE SOLUTION
Sodium alginate was dissolved in distilled water and stirred well to form a uniform solution.
5 MIXING ALL POWDERS INTO THE SOLUTION
All the prepared powders were added into the alginate solution and mixed thoroughly.
6 HOMOGENEOUS MIXTURE
A smooth and uniform mixture was obtained.
7 DRIPPING INTO CALCIUM CHLORIDE SOLUTION
The mixture was dropped slowly into calcium chloride solution.
8 BEAD FORMATION (CROSS-LINKING)
Calcium ions cross-linked with sodium alginate and formed stable beads.
9 WASHING
Beads were washed multiple times with distilled water to remove impurities.
10 DRYING
Beads were dried in sunlight or oven at low temperature until completely dry.
FINAL BEADS
(WITH ALL POWDERS)
The final beads contain moringa seed powder, banana peel powder, orange peel powder, charcoal powder and are ready to use for water purification.

ROADMAP:

GRINDING:

grinding and weighing of all powders.

sodium alginate dissolved in water (double boiling method).

ALGINATE GEL:

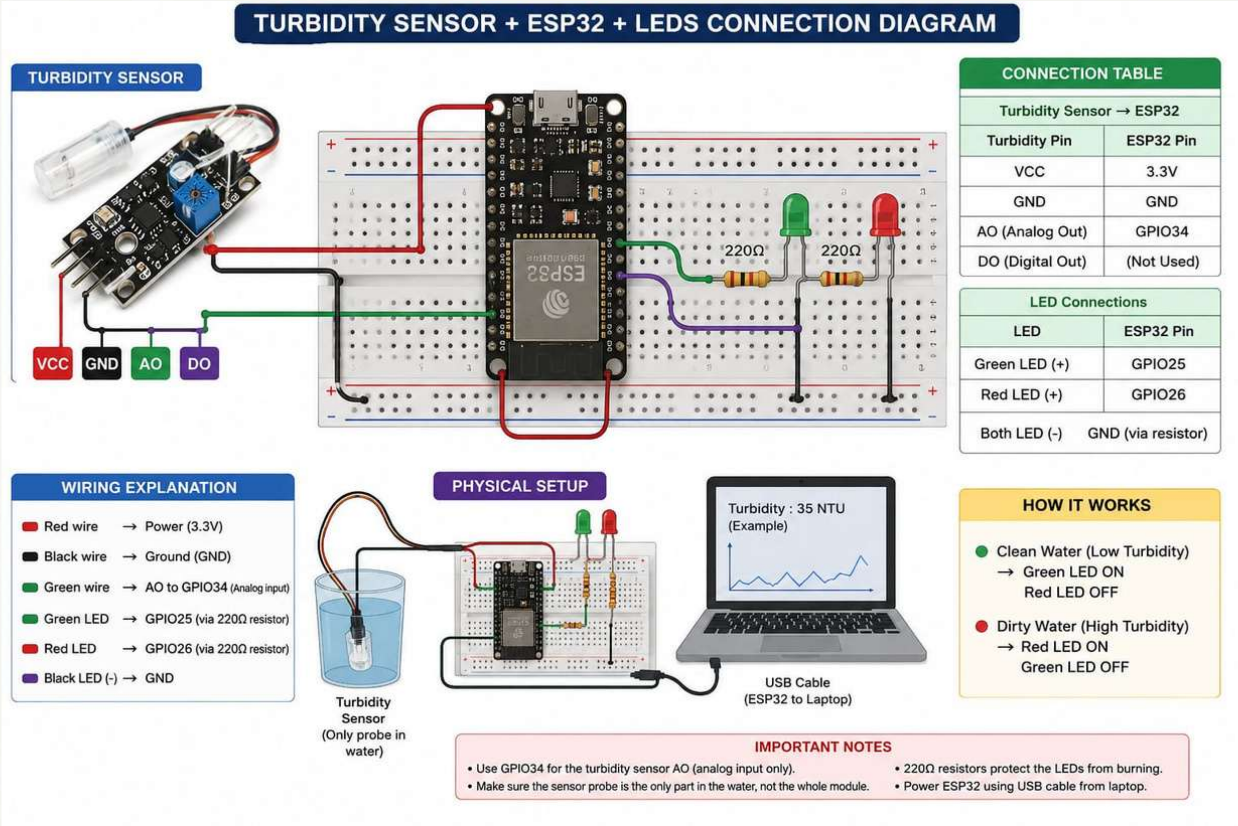
GEL MIXTURE:

mix the powders part by part into the gel.

drops were dropped into $CaCl_2$ using a syringe and dried.

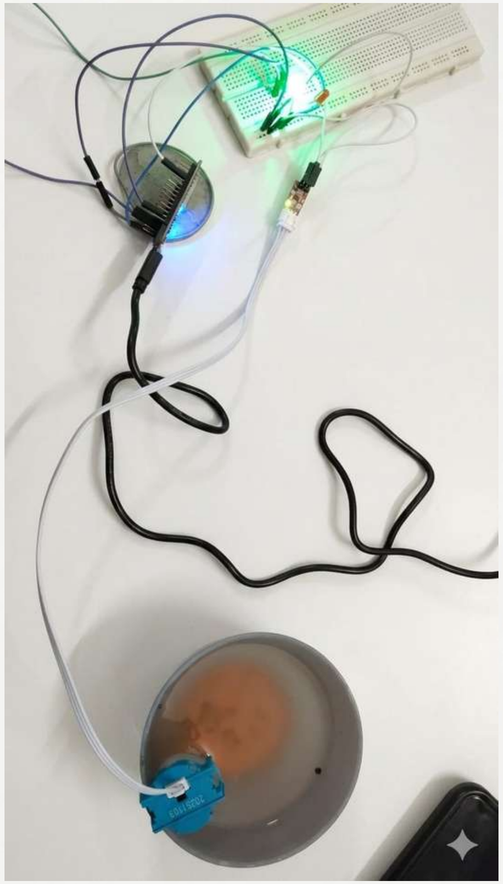
BEADS:

THE ET & ISE INTEGRATION



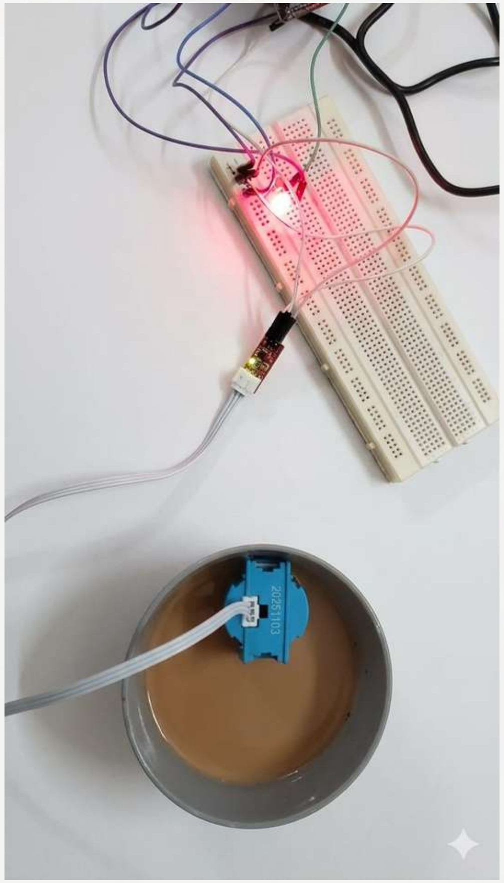
TURBIDITY MONITORING & NTU INTERPRETATION:

- **Sensor Principle:** sensor measures water cloudiness based on light scattering.
- **Data Processing:** ESP32 continuously reads values and converts them into Nephelometric Turbidity Units (NTU) for comparative analysis.
- **NTU VALUE INTERPRETATION:**
  - 0–50 NTU: Clear Water
  - 50–200 NTU: Slightly Turbid Water
  - 200–500 NTU: Moderately Turbid Water
  - Above 500 NTU: Highly Contaminated Water



SIGNIFICANCE OF SMART WATER MONITORING:

- **Continuous Oversight:** Enables consistent, real-time monitoring of water clarity providing instant feedback.
- **Rapid Data-Driven Decisions:** Facilitates instant comparative analysis between raw and treated water, ensuring treatment goals are met efficiently.
- **Validated Performance:** Smart integration confirms particle settling, floc formation, and true turbidity reduction with precise data points rather than subjective visual checks.



ADVANCING SUSTAINABLE SMART WATER ECOSYSTEM:

- **Low-Cost Implementation:** The use of **affordable** microcontrollers (like ESP32) and **accessible** turbidity sensors demonstrates smart water testing.

- **Sustainable Impact:**
  -Makes **natural**, **biodegradable** water clean-up methods practical and highly effective.
  -**Long-Term Protection**: Supports lasting, easy-to manage environmental monitoring.

ADVANTAGES OF OUR PROJECT:

- The major advantages of our project include:
    - Use of **biodegradable** and eco-friendly biomaterials.
    - Sustainable utilization of agricultural **waste products**.
    - Low-cost and **simple water treatment** approach.
    - Combination of **coagulation** and **adsorption** mechanisms within a single system.
    - **Reduced dependency** on chemical coagulants.
    - Smart **real-time turbidity** monitoring using ESP32 and sensors.

<u>FUTURE SCOPE:</u>

- Heavy **metal adsorption** analysis using advanced laboratory techniques.
- **Industrial wastewater** treatment applications.
- Real-time IoT **cloud monitoring** systems.
- Development of **large-scale purification** units.
- Regeneration and **reuse** studies of hydrogel **beads**.
- **AI** and data analytics integration for **smart water quality prediction**.
- Contaminated water can be made suitable for **drinking** as an upcoming project.

SDG RELEVANCE:

Helps improve access to **cleaner water**, supporting better **public health** and reducing water-related health risks.

The project promotes affordable and sustainable **water purification** methods for improving water quality.

The use of **agricultural waste** materials such as banana peel and orange peel supports **sustainable resource** utilization and waste reduction.

Development of **biodegradable** and **eco-friendly** purification systems helps reduce environmental pollution.

CONCLUSION:

This project demonstrates a sustainable and innovative approach toward water quality improvement using biodegradable materials and smart monitoring technology. The study highlights the potential of eco-friendly solutions in addressing environmental challenges and encourages further development in the field of smart water treatment systems.

THANK YOU!